The Effect of Contamination on Contamination Limit Regulation

US Coal Mining and Drinking Water Utilities

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Coal is not going away, and “clean” coal is not clean

- Coal supplied 16% of US electricity in 2025 (International Energy Agency (2025)).
- Federal policy is expanding mining access and delaying plant retirements (U.S. President (Trump) (2025); Daly (2025)).
- “Clean” coal $\equiv$ low sulfur coal $\equiv$ less acid rain.
- It says nothing about water: mining leaches arsenic, selenium, and barium into source water (Skousen, Ziemkiewicz, and McDonald (2019)).
- 85% of US households drink water from a utility that must treat and test that source water.

“Clean” coal $\equiv\ \leq$ 2% sulfur by weight.

The regulator only sees what the firm chooses to report

- Contamination limits, from taxes to drinking water, are policed by **self-reported compliance**.
- The regulator learns of an exceedance only if the firm tests and files.
- Two violations, very different prices:
  - **MCL** — reported contamination above the limit. Costly, often formally enforced.
  - **MR** — failure to monitor and report. Cheap.
- Rising contamination pushes both incentives the same way:
  - testing requirements, and so **reporting costs**, rise with impairment;
  - the **value of concealing** an exceedance rises too.
- If the price of silence does not rise with contamination, monitoring quietly fails.

Research question

What happens to firms and regulators when contamination of the regulated output rises exogenously, under self-reported compliance?

- Does contamination reduce firm self-reporting?
- Is the reduction concealment of exceedances, or avoidance of reporting costs?
- Does the regulator's enforcement rise to meet it?

What I do, and what is new

- Setting: community water systems (CWS) downstream of coal mines, 1985–2005.
  - Contamination arrives as an **upstream externality** — the firm cannot cut it by cutting output.
  - That separation is what makes the effect of contamination identifiable.
- Instrument: upstream coal sulfur % $\times$ post-1995, the Acid Rain Program Phase I deadline.
- Three things that are new:
  - First causal estimates of coal mining on **treated** drinking water at US utilities.
  - First evidence that **the cost of self-reporting**, not only concealment, drives water utility under-reporting (cf. Bennear, Jessoe, and Olmstead (2009); Zou (2021)).
  - First estimates of how **contamination itself** shifts regulator enforcement and inspection (Shimshack (2014)).

What I find

- $\uparrow$ mining $\rightarrow \uparrow$ contaminant concentrations. 10M cumulative tons upstream raises mean arsenic by 0.0023 mg/L, about 100% of its sample mean; selenium +69%, barium +24%. All stay below the MCL.
- $\uparrow$ mining $\rightarrow \uparrow$ **under-reporting**. One more upstream mine raises MR violations by 7–10 pp, more than doubling the base rate.
- $\uparrow$ mining $\rightarrow$ **no** change in MCL violations. The contamination record goes blank, not red.
- Costs, not concealment: when nitrate readings cross 50% of the MCL and monitoring requirements step up — exceedance risk fixed, reporting cost up — failure to report jumps.
- $\uparrow$ mining $\rightarrow \downarrow$ **formal enforcement** (-5.7 pp) but $\uparrow$ **sanitary visits** ($+14$ pp).

Contamination shifts monitoring from the firm onto the regulator, without ever raising the price of silence.

Setting and data

The SDWA runs on self-reports

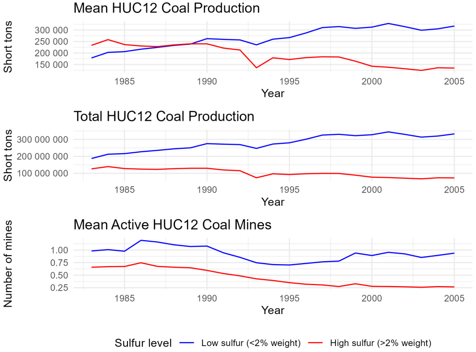
- Community water systems: $\geq$ 25 people, $\geq$ 15 connections, year-round.
- The utility tests its own water on a schedule and files results with the state.
  - Testing needs technical staff, administrative tracking, and paid laboratories.
  - Surveyed managers named reporting as the SDWA's single largest burden (Oleckno (1982)).
  - Schedules **tighten with assessed source-water impairment**.
- States hold primacy and wide discretion. Enforcement is remedial before punitive: informal (letters, notices, compliance plans), formal (consent decrees, fines, prosecution), and visits (sanitary, sampling, technical assistance).
- No self-report $\Rightarrow$ no MCL violation can be issued.

Mining puts inorganic chemicals in the water

- **Acid mine drainage**: exposed sulfide rock oxidises, forms sulfuric acid, dissolves metal ions and sulfates, which travel into surface and ground water (Skousen, Ziemkiewicz, and McDonald (2019)).
- **Coal wash slurry**: impurities stored in retaining ponds seep through porous rock, or the ponds fail.
- Both scale with the **sulfur content** of the coal seam.
- Contaminants: **arsenic, nitrate, barium, selenium** — all inorganic chemicals, all with MCLs unchanged over 1985–2005.
- Removing them is not free: replacing one small system's arsenic filtration cost $192,000 plus $60/day to run (US EPA, 2011).

The Acid Rain Program moved coal, not for water reasons

- 1990 Clean Air Act amendments cap SO_2 from power plants; Phase I deadline **1995**.
- Power plants were 90% of coal demand; switching to low sulfur coal was the main compliance response.
- Demand shifts **toward low sulfur**, **away from high sulfur** seams.
- High sulfur mines cannot wash the sulfur out — it is bound into the coal's chemical structure.



Model: the utility's two decisions

Utility draws compliance cost $\theta \sim F(\cdot\,;m)$ and reporting cost $c \sim G(\cdot\,;m)$, both worsening in upstream contamination m.

- Chooses negligence a, unobserved, with benefit $\theta b(a)$, $b' > 0$, $b'' < 0$.
- Each of N required tests would reveal an exceedance with probability $q(a)$, $q' > 0$.
- Penalties: r for a self-reported exceedance, t for an MR violation, $s > r$ if an audit (probability p) uncovers a hidden exceedance.

$$\underbrace{c + q(a)\,r}_{\text{report}} \lesseqgtr \underbrace{t + p\,q(a)\,s}_{\text{stay silent}} \implies \text{report iff } c \le \hat{c}(a) \equiv t - (r - ps)\,q(a).$$

- A system with **zero** exceedance risk still stops reporting once $c > t$.
- Silence is cheap at the margin: $ps < r$.

Model: why the utility goes quiet

Proposition 1

Negligence rises in compliance cost: $\partial a / \partial \theta > 0$ within each region.

Proposition 2

Under-reporting rises with contamination, $\partial MR / \partial m \geq 0$, through **(A) cost** — reporting costs shift up against a fixed cutoff — and **(B) concealment** — the population shifts toward high-negligence types whose cutoff is already lower.

Corollary 1

Hold exceedance risk fixed, $q(a_{SR}(\theta)) \equiv \bar{q}$. The cutoff is then type-invariant, Channel B integrates to zero, and *all* of $\partial MR / \partial m = -\,\partial G(\bar{c}\,; m)/\partial m \geq 0$ runs through Channel A.

Concealment needs exceedance risk to move; cost does not. Hold risk fixed, let reporting costs jump — if under-reporting still rises, it is cost, not hiding.

Model: the regulator has a lever, and it is not the limit

Proposition 3

Scaling all penalties by $\lambda > 0$ lowers negligence and raises the reporting threshold, $C^*(\lambda) = \lambda\,\hat{c}(a_\lambda)$. Reporting costs are technology, not legal penalties, so they do not scale with enforcement.

- Under-reporting driven by Channel A is **correctable**: raising the price of silence pulls the cutoff back up above the risen reporting cost.
- Tightening the MCL does not. That moves exceedance risk — under Corollary 1, precisely the margin that is not doing the work.
- So the question for the regulator is not whether it *could* respond, but whether it *does*.

Proposition 2 and Corollary 1 are the test of the mechanism. Proposition 3 says the lever is the price of silence, which makes the observed enforcement response the thing to measure.

Data

- **Violations, enforcement, visits** — SDWIS/ECHO, universe of CWSs 1985–2005. Binary by CWS $\times$ year; days-in-year as a robustness measure.
  - Under-inclusive but accurate: 95% of the violations SDWIS holds are correctly reported (US EPA, 2000).
- **Contaminant concentrations** — EPA Six-Year Review round 2, 1998–2005. Cleaned following Ravalli et al. (2022); non-detects set to MDL/$\sqrt{2}$.
- **Mines** — MSHA locations merged to EIA annual production, mine by mine.
- **Coal sulfur %** — USGS COALQUAL cores, averaged within 20 km of each watershed.
- **Hydrology** — NHDPlus HUC12 boundaries and inflow/outflow.
- Sample: CWSs with an intake **strictly one HUC12 downstream** of a mine watershed. 6,225 CWS-years, 333 CWSs.

Matching intakes to the watershed above them

- HUC12 is the finest watershed mapped for the continental US (>87,000 units).
- Exposure = mines and tons in the HUC12s **directly upstream** of the CWS's intakes.
- Intakes **colocated** with a mine are excluded: an intake at a watershed's inflow need not be downstream of a mine at its outflow.
- Multiple intakes: sum production, average sulfur.

MR violations are common; MCL violations barely exist

Panel A: Any violation in year				
	MR		**MCL**	
Contaminant	**%**	**N**	**%**	**N**
Nitrates	6.05	377	0.06	4
Arsenic	4.01	250	0.03	2
IOCs	4.59	286	0.32	20

Panel B: Days in year								
	MR				**MCL**			
Contaminant	**Mean**	**SD**	**P90**	**P99**	**Mean**	**SD**	**P90**	**P99**
Nitrates	15.53	67.32	0.00	365.00	0.18	8.01	0.00	0.00
Arsenic	10.69	56.01	0.00	365.00	0.04	2.59	0.00	0.00
IOCs	12.27	60.15	0.00	365.00	0.78	15.83	0.00	0.00

Nitrate MR 6.1%, arsenic 4.0%, IOC 4.6% of CWS-years. MCL violations: 4, 2, and 20 events across twenty years.

Does coal mining contaminate utility water?

Design: cumulative production, measured concentrations

$$Concen_{cht} = \beta \cdot \sum_{\tau=1985}^{t} \text{UpstreamCoalProd}_{c\tau} + \rho_c + HUC02_h \cdot \tau_t + \epsilon_{cht}$$

- $Concen_{cht}$: mean annual concentration at CWS c in basin h, year t (SYR2, 1998–2005).
- β: effect of 10 million additional cumulative short tons mined upstream since 1985.
- ρ_c: CWS fixed effects. $HUC02_h \cdot \tau_t$: basin-specific year effects, absorbing region-wide industrial trends.
- SEs clustered at the CWS level.

- Continuous treatment, **no pure control**: identification comes from relative exposure.
- Assumption: CWSs with large upstream production changes would have trended like those with small ones.

Mining raises contaminant levels — but not above the limit

	Mean conc.			
	Arsenic	Nitrate	Barium	Selenium
	(1)	(2)	(3)	(4)
Cumul. upstream coal prod. (10M ST)	0.0023***	0.0572	0.0171*	0.0033**
	(0.0004)	(0.0922)	(0.0093)	(0.0016)
Observations	334	432	322	319
Utility fixed effects	✓	✓	✓	✓
huc02-year fixed effects	✓	✓	✓	✓

Notes: Standard errors clustered at the utility level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Arsenic +0.0023 mg/L per 10M tons: about 100% of its mean, and 4.6% of the pre-2006 MCL. Selenium +69% and barium +24% of their means. Nitrate is positive but imprecise. **Every reading in the sample stays under its MCL.**

Do utilities keep self-reporting?

IV design: the ARP as a shifter of where coal is mined

First stage:

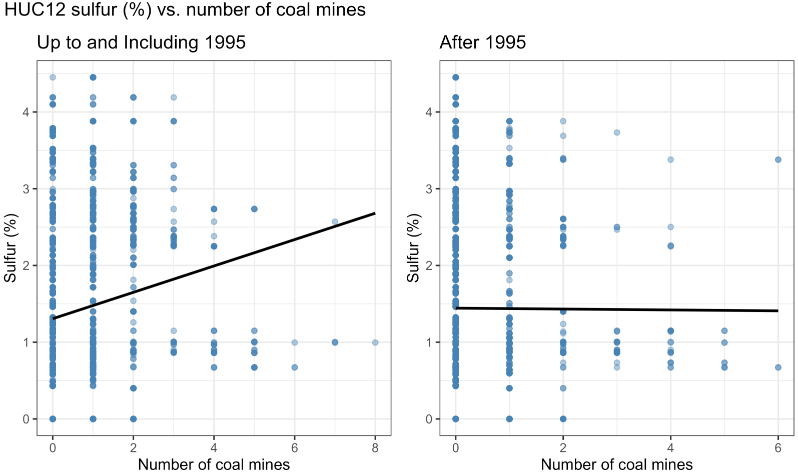
$$\begin{aligned}\mathsf{NumMines}_{ct} = {} & \alpha_0 + \alpha_1 \mathsf{NumFacility}_{ct} + \alpha_2 \mathsf{UpstreamCoalSulfur\%}_c \cdot \mathsf{Post95}_t \\ & + Y_t + C_c + \nu_{ct}\end{aligned}$$

Second stage:

$$\mathsf{Violation}_{vct} = \beta_0 + \beta_1 \mathsf{NumFacility}_{ct} + \gamma \widehat{\mathsf{NumMines}}_{ct} + Y_t + C_c + \epsilon_{ct}$$

- $\mathsf{Post95}_t = \mathbf{1}\{t > 1995\}$. Sulfur is time-invariant, so C_c absorbs its level.
- Outcome: 1 if CWS c had a violation of type v at any point in year t. Coefficients are in percentage points.
- SEs clustered at the CWS level.

Before 1995 mines followed sulfur; after 1995 they did not



Sample: mine HUC12s (D1) upstream of downstream-only 2SLS CWS intakes, >= 1 active mine year 1985-2005.

First stage: high sulfur watersheds lost their mines

	Upstream coal mines (sum)
	(1)
post95 $\times$ Upstream sulfur %	-0.2404***
	(0.0458)
F-test (1st stage, clustered)	27.52
Observations	6,225

Notes: All specifications include CWS and year fixed effects. Standard errors clustered at the CWS level. *** p<0.01, ** p<0.05, * p<0.1.

$F = 27.5$, above the 23.1 critical value of Montiel Olea and Pflueger (2013) for a 5% test against 10% worst-case bias, and well above the rule of thumb of 10 (Staiger and Stock (1997)).

Why I believe the instrument

- **Independence.** The ARP is national clean-*air* law. Its scope, stringency, and 1995 date were set by national energy and emissions politics, not by any downstream utility's water quality or its influence over mine siting. Time-invariant sulfur exposure is absorbed by CWS fixed effects.
- **Exclusion.** The threat is a post-1995 SDWA change hitting high- and low-sulfur utilities differently. National rule changes are absorbed by year effects. Directly testable: regress the number of intake facilities — the observable proxy for the model's negligence type — on the instrument. The coefficient is -0.0016 (s.e. 0.0016), with and without a balanced panel.
- **Monotonicity.** A defier is a high-sulfur watershed that expanded *because* its coal was high sulfur. Scrubber adopters lose share rather than gain it; low-cost high-sulfur regions that expand anyway are never-takers. Monotonicity is pairwise in sulfur, so low- and high-sulfur areas may respond in opposite directions (Angrist and Imbens (1994)).

Mining makes utilities stop reporting

	Nitrates (MR)	Arsenic (MR)	Inorganic chemicals (MR)
Panel A: OLS			
Upstream coal mines (sum)	1.95***	1.70***	1.80***
	(0.72)	(0.65)	(0.66)
Panel B: Reduced Form			
post95 × Upstream sulfur %	-2.39***	-1.79**	-1.70**
	(0.84)	(0.73)	(0.77)
Panel C: 2SLS			
Upstream coal mines (sum)	9.93***	7.43**	7.07**
	(3.73)	(3.05)	(3.23)

Notes: All specifications include utilities and year fixed effects. Standard errors clustered at the utilities level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

One more upstream mine: nitrate MR +9.9 pp, arsenic +7.4 pp, IOC +7.1 pp, against base rates of 6.1%, 4.0%, and 4.6%. OLS is roughly five times smaller — negative omitted variable bias, as expected if utilities that can keep mines out are also better at compliance.

And the contamination record goes blank, not red

	Nitrates (MCL)	Arsenic (MCL)	Inorganic chemicals (MCL)
Panel A: OLS			
Upstream coal mines (sum)	-0.02	-0.02	-0.00
	(0.02)	(0.04)	(0.14)
Panel B: Reduced Form			
post95 $\times$ Upstream sulfur %	0.00	-0.02	0.19
	(0.00)	(0.01)	(0.16)
Panel C: 2SLS			
Upstream coal mines (sum)	-0.02	0.08	-0.78
	(0.02)	(0.06)	(0.69)

Notes: All specifications include utilities and year fixed effects. Standard errors clustered at the utilities level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

MCL effects are economically and statistically indistinguishable from zero. This is not evidence that the water stayed clean: without a self-report, the regulator cannot issue an MCL violation at all. It is also the wrong prediction for concealment — hiding is worth paying for when an exceedance is at stake, and measured concentrations never get there.

Reporting cost, not concealment: Corollary 1 in the data

- Nitrate rules require **more frequent monitoring** once a reading exceeds 50% of the MCL — a discrete jump in c, with no jump in the penalty for silence and no jump in exceedance risk: a reading at half the limit is still a reading under the limit.
- That is Corollary 1's condition, met in the field. Exceedance risk is held fixed, reporting cost shifts up, Channel B is switched off — so whatever happens to under-reporting here is Channel A.

	Arsenic MR (1-yr)	Arsenic MR (6-mon)	Nitrate MR (1-yr)	Nitrate MR (6-mon)	Pooled IOC (1-yr)	Pooled IOC (6-mon)
	(1)	(2)	(3)	(4)	(5)	(6)
Mean concen. (z-score)	0.2622	0.2665	1.2804	0.0482		
	(0.3250)	(0.2495)	(1.4274)	(0.8741)		
Concen. > 50% MCL			58.9693**	26.1081**	0.4033	-2.1966
			(24.6040)	(10.5298)	(2.4182)	(1.4172)
Concen./MCL					-5.9684	-5.7062
					(8.1107)	(4.0256)
Observations	419	419	851	851	3,705	3,705
CWS fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓	✓	✓	✓
Contaminant fixed effects					✓	✓

Notes: SEs clustered at the CWS level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Crossing 50% of the nitrate MCL raises the probability of failing to report within six months by 26 pp, and within a year by 59 pp. Descriptive, not causal — read the sign, not the magnitude. Utilities go quiet exactly where reporting gets expensive, not where there is something to hide.

How do regulators respond?

Enforcement moves the wrong way

	Any informal enforcement	Any formal enforcement	No enforcement
Panel A: OLS			
Upstream coal mines (sum)	0.84	-1.69***	-0.78
	(0.79)	(0.45)	(0.86)
Panel B: Reduced Form			
post95 × Upstream sulfur %	0.14	1.45***	0.03
	(1.02)	(0.52)	(1.08)
Panel C: 2SLS			
Upstream coal mines (sum)	-0.56	-5.65***	-0.10
	(4.00)	(2.05)	(4.23)

Notes: All specifications include utilities and year fixed effects. SEs clustered at the utilities level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

One more upstream mine cuts formal enforcement by 5.7 pp against a base rate of 3.5%; informal enforcement is flat. Formal action needs evidence, and under-reporting is precisely the absence of evidence.

The regulator substitutes its own monitoring instead

	Sanitary visits	Technical assistance	Enforcement visits	Sample collection	Inspection
Panel A: OLS					
Upstream coal mines (sum)	1.06	-0.38	-0.71**	-0.19	-0.33
	(0.67)	(0.42)	(0.32)	(0.12)	(0.47)
Panel B: Reduced Form					
post95 $\times$ Upstream sulfur %	- 3.60***	-0.36	-0.94**	-0.34	0.48
	(1.03)	(0.48)	(0.40)	(0.27)	(0.60)
Panel C: 2SLS					
Upstream coal mines (sum)	14.06***	1.41	3.67*	1.35	-1.88
	(4.87)	(1.96)	(1.88)	(1.15)	(2.41)

Notes: All specifications include utilities and year fixed effects. SEs clustered at the utilities level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Sanitary visits +14.1 pp against a base rate of 14.6%; enforcement visits +3.7 pp. Regulators show up — and then do not sanction. Either there are no hidden exceedances, or visits do not find them (Mi, Zhang, and Liu (2026)); I cannot separate the two. Either way the price of silence never moves, which is the one thing Proposition 3 says would work.

Conclusion

- Exogenous contamination raises the cost of self-reporting, and self-reporting falls — **even where the quality standard is never breached**.
- The evidence points to cost avoidance, not concealment. Corollary 1 makes the two separable: with exceedance risk held fixed, only the cost channel can move under-reporting — and that is exactly where reporting failures spike. Concentrations stay below MCLs and regulators visit but do not prosecute, both of which cut against a hiding story.
- Enforcement does not keep pace. Formal action falls as contamination rises, and the regulator absorbs the monitoring burden through sanitary visits instead.
- **Policy.** Tightening MCLs does nothing if nobody tests — it moves the one margin Corollary 1 rules out. By Proposition 3 the binding lever is the price of *not* reporting where source water is impaired, and it is not currently being pulled (Malik (1993); Harford (1987)).
- Open question: whether shifting monitoring onto the regulator is efficient depends on relative monitoring costs and on regulator preferences.

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Appendix

Appendix: combined MR and MCL violations

	Nitrates	Arsenic	Inorganic chemicals
Panel A: OLS			
Upstream coal mines (sum)	1.94***	1.67**	1.75***
	(0.72)	(0.65)	(0.65)
Panel B: Reduced Form			
post95 × Upstream sulfur %	-2.38***	-1.80**	-1.52*
	(0.84)	(0.73)	(0.80)
Panel C: 2SLS			
Upstream coal mines (sum)	9.91***	7.50**	6.34*
	(3.72)	(3.06)	(3.32)

Notes: All specifications include utilities and year fixed effects. Standard errors clustered at the utilities level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

The combined effect is the MR effect: nitrate +9.9 pp, arsenic +7.5 pp, IOC +6.3 pp.

Appendix: instrument balance on intake facilities

	Number of Intake Facilities	
	(1)	(2)
Upstream sulfur		0.0384
		(0.0735)
Post-1995 $\times$ Upstream sulfur	-0.0016	-0.0016
	(0.0016)	(0.0017)
Utility fixed effects	✓	
Year fixed effects	✓	✓
Balanced panel		✓
Utilities	333	260
Observations	6,225	5,460

Notes: Standard errors, clustered at the utility level, are shown in parentheses below each coefficient. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Appendix: concentration summary statistics

Variable	MCL	Mean	Max	Std. Dev.	N	Near MCL
Panel A: Full sample						
Arsenic (mg/L)	Pre-2006: 0.050 mg/L, 2006+: 0.010 mg/L	0.0029	0.0110	0.0023	378	0.0%
Nitrate (mg/L)	10.0 mg/L	0.7520	5.5400	0.8482	444	0.5%
Barium (mg/L)	2.000 mg/L	0.0748	0.7071	0.0671	369	0.0%
Selenium (mg/L)	0.050 mg/L	0.0048	0.0354	0.0051	367	0.5%
Cumul. upstream coal prod. (10M ST)	—	0.4402	14.1792	1.4162	518	—
Panel B: Utility exposed to above median cumulative tons of coal extraction						
Arsenic (mg/L)	Pre-2006: 0.050 mg/L, 2006+: 0.010 mg/L	0.0031	0.0100	0.0026	177	0.0%
Nitrate (mg/L)	10.0 mg/L	0.7770	5.5400	0.8175	236	0.4%
Barium (mg/L)	2.000 mg/L	0.0754	0.3320	0.0579	169	0.0%
Selenium (mg/L)	0.050 mg/L	0.0040	0.0346	0.0043	169	0.6%
Cumul. upstream coal prod. (10M ST)	—	0.8822	14.1792	1.9089	258	—

Appendix: who appears in the Six-Year Review

	(1) Has SYR2 reading	(2) No SYR2 reading	Diff. (1)−(2)	p-value
Ever MR violation, 1997–2005	32.4%	9.8%	22.6%***	0.000
Ever MR violation, 1985–2005	51.1%	32.8%	18.3%***	0.001
Ever MCL violation, 1997–2005	1.3%	0.0%	1.3%*	0.083
Ever MCL violation, 1985–2005	5.3%	1.6%	3.7%*	0.052
Mean annual violations (count, 0–6), 1997–2005	0.09	0.11	-0.02	0.663

Utilities with SYR2 readings are 18–23 pp more likely to ever incur an MR violation. The concentration sample over-represents the more negligent utilities, so it is not selected toward clean systems.

Appendix: enforcement and visits in the panel

	Whole panel		During MR year		During MCL year	
	%	**N**	**%**	**N**	**%**	**N**
Panel A: Enforcement type						
Formal	3.50	218	9.53	43	7.69	2
Informal	19.02	1,184	67.63	305	53.85	14
Any	21.37	1,330	70.73	319	65.38	17
Panel B: Visit type						
Sanitary	14.55	906	14.19	64	7.69	2
Technical assistance	2.33	145	4.88	22	0.00	0
Enforcement	1.16	72	2.44	11	3.85	1
Sample collection	0.90	56	0.44	2	0.00	0
Inspection	4.05	252	5.76	26	3.85	1

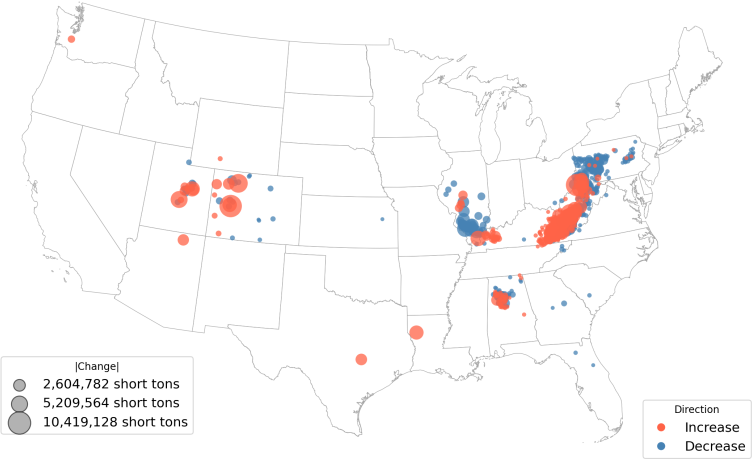
Enforcement responds to violations — any enforcement rises from 21% overall to 71% in MR years — but sanitary visits are no more likely in MR years than in any other year.

Appendix: reporting lag, national downstream states

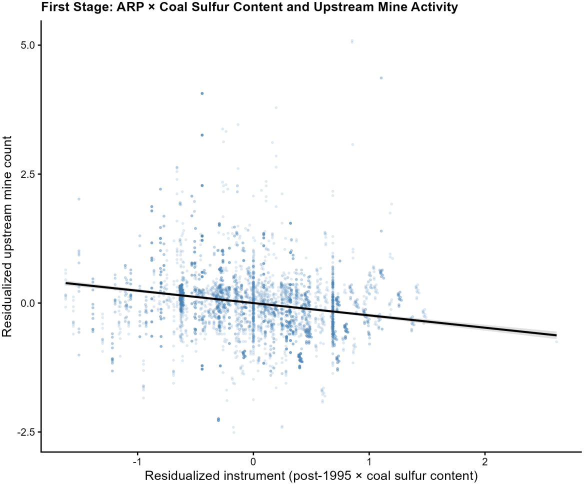
	Nitrate MR (1-yr)	Nitrate MR (6-mon)	Nitrate MR (1-yr) Logit	Nitrate MR (6-mon) Logit
	(1)	(2)	(3)	(4)
	OLS	OLS	Logit	Logit
Concen. $>$ 50% MCL	0.4226*	0.1529*	0.1568**	0.0969
	(0.2498)	(0.0840)	(0.0761)	(0.0676)
Mean concen. (z-score)	2.1957**	1.0630***	0.5452***	0.3686***
	(0.9297)	(0.3789)	(0.1906)	(0.1327)
Observations	541,483	541,483	86,940	65,317

Notes: All specifications include CWS and year fixed effects. SEs clustered at the CWS level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

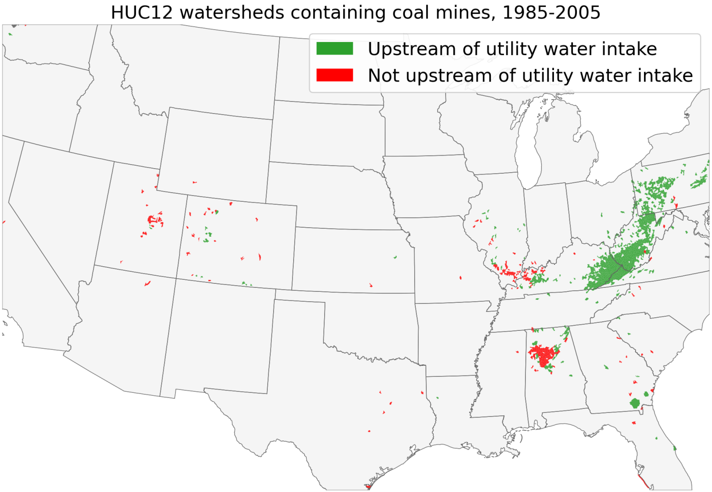
Appendix: where coal production moved, 1985–2005



Appendix: residualized first stage



Appendix: sample watersheds



Appendix: Corollary 1, statement and proof

Write the share of types under-reporting as $h(\theta, m) = 1 - G(c^*(\theta); m)$, with cutoff $c^*(\theta) = t - (r - ps)\, q(a_{SR}(\theta))$, so that

$$\frac{\partial MR}{\partial m} = \underbrace{\int_0^{\bar\theta} \frac{\partial h(\theta, m)}{\partial m}\, f(\theta; m)\, d\theta}_{\text{A: cost}} + \underbrace{\int_0^{\bar\theta} h(\theta, m)\, \frac{\partial f(\theta; m)}{\partial m}\, d\theta}_{\text{B: composition}}.$$

Corollary 1. If $q(a_{SR}(\theta)) \equiv \bar q$ on $(0, \bar\theta)$, then $\partial MR/\partial m = -\,\partial G(\bar c; m)/\partial m \geq 0$, where $\bar c = t - (r - ps)\bar q$.

Proof. The cutoff is then type-invariant, $c^*(\theta) \equiv \bar c$, so $h(\theta, m) = 1 - G(\bar c; m)$ does not vary with θ and comes outside both integrals. The limits 0 and $\bar\theta$ do not depend on m, so by Leibniz's rule

$$\mathrm{B} = \left[1 - G(\bar c; m)\right] \frac{d}{dm}\left[\int_0^{\bar\theta} f(\theta; m)\, d\theta\right] = \left[1 - G(\bar c; m)\right] \frac{d}{dm}[1] = 0,$$

since $f(\cdot\,; m)$ is a density for every m. Channel A collapses to $-\,\partial G(\bar c; m)/\partial m \geq 0$, because contamination shifts reporting costs up. $\square$

Why it matters. Channel B is the concealment margin: it operates only if contamination reallocates mass toward types whose exceedance risk, and hence whose cutoff, differs. Shut it down and any remaining rise in under-reporting is avoidance of the reporting technology itself. The 50%-of-MCL nitrate step is the empirical counterpart — monitoring frequency jumps, the sanction schedule does not, and readings are nowhere near the limit.

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